

General-regime watch: resolved quantitatively (2026-10-06)

Questions and method in the **protocol** (README.md). Rounds 1-3: thin-product interface (merged PR #1485), twelve-target survey (open PR #1493), refutation audit (open PR #1494). Status labels: **[proposed]** / **[derived]** / **[verified]** / **[measured]** (none in this round).

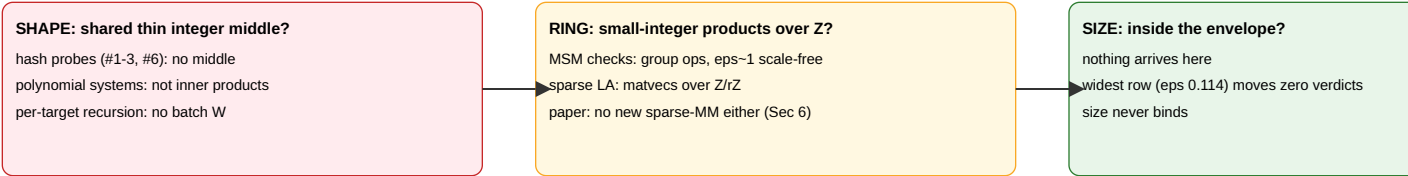
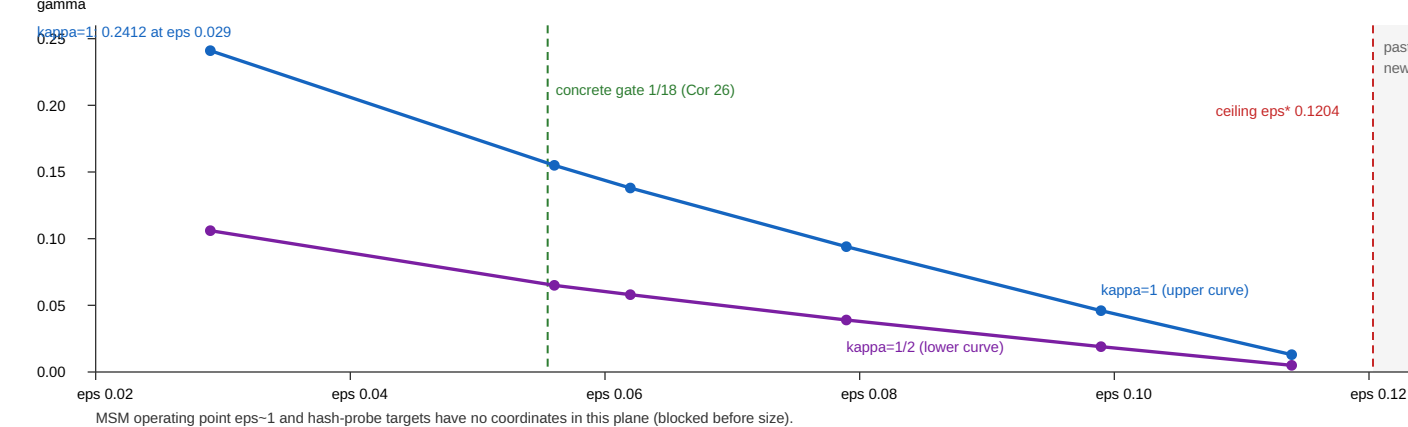
Headline

The envelope widens; nothing moves. The proven regime runs from the concrete line ($D \leq N^{(1/18)}$, $\gamma 0.063$) out to $D \leq N^{0.114}$ with γ up to 0.2412 for very thin, very sparse wants (frozen in table2.json, verified against a render of paper p.40). All sixteen targets stay blocked at shape, ring, or scope: **size never binds**, so widening size changes no verdict.

Envelope and funnel

The general envelope widens size -- and size never binds

Proven $\gamma(\text{eps})$ from paper Table 2 (Thm 25), plotted from table2.json. All 16 repo targets stop at shape/ring/scope.



Watch condition (conjunction): a new base identity past $\text{eps}^* = 0.1204$ AND a reduction reshaping an algebraic workload into thin products. Either alone changes nothing.
Gate-parameterization next action: extend screen_workload with the six table2.json rows after PR #1493 merges (acceptance: existing tests unchanged).

Editable source: figure.mmd. Envelope data: table2.json (paper Table 2, p.40, cells verified against rendered page).

Figure 1 -- proven $\gamma(\text{eps})$ envelope from paper Table 2 (curves plotted from table2.json), and where the sixteen screened targets stop. Green size gate: nothing arrives. Editable source: figure.mmd.

Re-screen (widest proven row: eps 0.114, kappa up to 1)

Target group	Blocking gate [derived]	Envelope effect
4 insertion points (relation batch, factor queries, sparse LA, descent)	Shape/ring/model	None : no (D, W) mapping to resize
Pair-table / MITM / BSGS / rho probes; Macaulay (#1-3, #6)	Shape: hash probes, no middle	None : no plane coordinates
Batch MSM (#4); isogeny-route search (#10)	Ring, then scale-free size (eps ~ 1)	None : eps ~ 1 off every row
3SUM/3XOR transfers (#7-8)	Missing cross-domain reduction	None : envelope is not a reduction
Convolution, real-RAM, SETH/OV (#9, #11-12)	Out of scope	None

Sparse multiplication, reduction losses [derived]

- Sparse MM closed twice over.** The paper (Sec. 6): its techniques "do not seem to give new algorithms for sparse matrix multiplication" (balanced case untouched). Ours stacks on top: block Wiedemann is sparse *matvecs* over $\mathbb{Z}/r\mathbb{Z}$ -- wrong operation and

wrong ring. Greps for girth, sparse-MM hardness, and balanced-triangle vocabulary return nothing outside our own study files.

- **Kept saving is the metric.** Hardness reductions reused as algorithms keep only a fraction of the exponent saving (half, a third); SVX27 black-box optimality forces tighter transfers to use problem structure. That is our 5 rule in other words: end-to-end kept savings only, never headline exponents.
- **Gray boxes mirror the audit.** Merely 3SUM/APSP-hard problems (reductions going the wrong way) get no speedup -- consistent with round 3's unaffected classifications.

Standing watch condition [proposed]

The technique family reaches our workloads only through **two** independent breakthroughs: (a) a new base identity lifting the ceiling past $\epsilon^* = 0.1204$ (the paper's open computer-search problem), **and** (b) a reduction reshaping an algebraic IC workload into thin small-integer products with known sparse W . Either alone changes nothing. Next action: parameterize `screen_workload` on the six `table2.json` rows after PR #1493 merges (existing tests unchanged).

Boundary, table, ratio

- Boundaries unchanged: generic floor + matched ρ , unit S .
- No table row produced: envelope cells are the paper's transcribed numbers; re-screen verdicts are gate evaluations, kept out of every result panel.
- Falsification target: a workload whose only block is size under the concrete line but passes a general row. The table above is the search that found none.
- All phases unpriced (`null`); wall time appears nowhere.

Graphs checked -- no change [verified by inspection]

- Scoreboard + progress-timeline: no new ratio, no point added.
- Leaderboard files, browser data, curve registry: nothing landed.
- Figure 1 is a new visual for this round (envelope plotted from frozen data), not a canonical-graph update.

Sources

- Paper: <https://arxiv.org/pdf/2610.06783> (v1); Table 2 + Theorems 24-25 (p.40); Sec. 6.
- Frozen envelope: `table2.json` (this directory).
- Rounds 1-3: `research/lopsided_thin_product_20261006/`, `research/lopsided_other_speedups_20261006/`, `research/lopsided_implications_20261006/`.